Transcutaneous Spinal Cord Stimulation-Augmented Treadmill Gait Training in Chronic Stroke: Single-Session and Preliminary Longitudinal Effects on Step Length

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Abstract—**Transcutaneous spinal cord stimulation (tSCS) is a safe and feasible approach to augment traditional therapies. Here, we explore the use of tSCS to augment treadmill gait training in people with chronic stroke. Building on our prior tSCS work in this population, we describe step length and step length symmetry responses to single-day and longitudinal tSCS-augmented, symmetry-focused treadmill training.**

We report small, clinically insignificant single-day step length increases after one bout of treadmill training, and no evidence of a universal "best" single-day tSCS parameters. Longitudinal training increased step length, via the non-paretic side or bilaterally, for all participants. Step length symmetry improved in only one person.

We preliminarily show that longitudinal tSCS-augmented treadmill training increased step length after 12 training sessions. We are actively recruiting to expand the longitudinal sample size, including a control group. Future work should also explore the effect of tSCS on additional gait metrics.

I. Introduction

TRANSCUTANEOUS spinal cord stimulation (tSCS) is a promising neuromodulation technique to facilitate motor recovery whereby electrical stimulation is applied via an electrode placed over the skin of the spinal cord. Originally formulated for use in people with spinal cord injury (SCI), evidence of physiological changes in the spinal cord post-stroke [1] motivated its use in stroke patients.

Best tSCS practices in the SCI population may not translate to people with stroke. Our lab has previously performed a series of experiments to understand how participant demographics [2] and tSCS parameters [3] affect comfort and muscle responsiveness in chronic stroke. However, the effect of a single session or longitudinal tSCS-augmented gait training to improve post-stroke gait symmetry is unknown, as is the effect of tSCS parameter selection.

The purpose of this study is to quantify the acute effects of tSCS on gait symmetry in a chronic stroke cohort. We also report initial findings of the effects of 8 weeks of longitudinal treadmill gait training in four of the same people.

II. Material and Methods

Here we report on a two-part study, approved by the Northwestern University Institutional Review Board. In the first part, 20 chronic stroke patients (13 m, 7 f; 53.3 ± 10.8 yrs of age; 4.3 ± 2.5 yrs post-stroke) completed six sessions of 25 minutes intervention of moderate intensity treadmill gait training plus tSCS. In two sessions, they received SHAM stimulation (0 mA; first SHAM removed from analysis as a wash-in). In the other sessions, tSCS was administered with a unique combination of stimulation intensity and burst frequency (stimulation intensity at participant tolerance (TOL) or 30% of the lowest resting motor threshold across muscles (RMT_{min}), burst frequency 30 or 50 Hz). RMT's were assessed via spinal motor evoked response testing. In all sessions, we measured step length using GAITRite (GAITRite, Franklin NJ USA) and computed step length symmetry (0-200 scale, 0 is perfect symmetry) change PRE to POST intervention using Cohen's d effect size to quantify single-session responses to tSCS. Two-sample paired t-tests on average tSCS vs. SHAM assess single-day effects. tSCS parameter effects are assessed via one-way RMANOVA on the Cohen's d values.

In an ongoing second part of the study – a longitudinal, double-blind clinical trial – four chronic stroke patients (4 m 0 f; 49.8 ± 10.4 yrs of age; 5.6 ± 3.6 yrs post-stroke) of a planned 20 have each completed between 12 and 24 of 24 total intervention sessions. Each session consists of 30 minutes of moderate intensity tSCS-augmented treadmill gait training at an average frequency of three sessions per week. tSCS intensity is set to patient tolerance, with 30Hz burst frequency. Step length and step length symmetry were quantified every three training (TX) sessions, and at baseline (BL), after 12 sessions (MID), after 24 sessions (POST), and at one- and three-months follow-up (1MO & 3MO). A one-way ANOVA for each subject with factor *Timepoint* assesses differences across sessions. As the four currently enrolled participants are all in the tSCS group, no *Group* analysis is presented here. For both study parts, all analyses are conducted on fast-velocity gait.

III. Results

A. Single-Day Effects

We observed a statistically significant group-level improvement in step length in tSCS vs. SHAM (mean Δ step length ~2 cm; Δ Cohen's d = 0.14 ± 0.26, p=0.02) after a single 25-minute bout of tSCS-augmented treadmill training (Fig. 1A), but not in step length symmetry (Δ Cohen's d =

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0.06 ± 0.61, p=0.69). We found no difference between any of the four stimulation parameters in step length (p=0.34) or step length symmetry effect sizes (p=0.20).

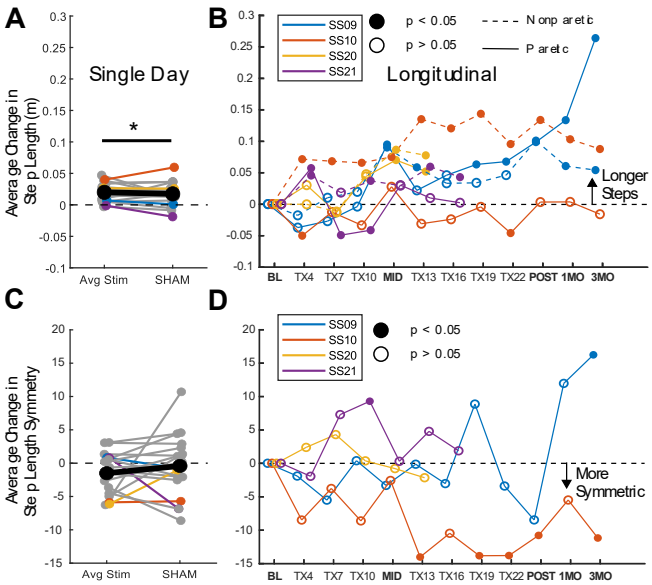

Fig. 1. Change in step length (A, B) and step length symmetry (C, D) in each session of the single-day (A, C) and longitudinal (B, D) study parts. Colors indicate the same people in both study parts. Black is the group mean of 20 people (A, C). In (B, D), significant differences from baseline are indicated by filled circles. "TX" sessions are every third treadmill training session. Assessment sessions are bolded: baseline (BL), midpoint (MID), post (POST), and follow-ups (1MO, 3MO).

B. *Longitudinal Training Effects*

We found a significant main effect of *Timepoint* on step length for all participants ($p<0.0001$), with significant changes from baseline (filled circles, Fig. 1B). One person significantly improved step length symmetry (SS10). The rest showed no change (SS20) or became less symmetric at one timepoint each (SS09, SS21).

IV. Discussion

This two-part study explores the effects of and relationship between single-day and longitudinal tSCS-augmented treadmill training on step length and step length symmetry. From PRE- to POST- one 25-minute treadmill training session, step length statistically significantly increased using tSCS more than SHAM, though this change may not be practically significant (~2 cm). No single-day difference in step length symmetry was found. Longitudinal sessions showed person-specific step length improvements and significant step length symmetry changes in only one person.

A. *Single-Day Effects*

Most people slightly increased step length after a single session of 25 minutes of treadmill training (Fig. 1A) but on average did not change step length symmetry (Fig. 1C), and tSCS did not elicit clinically superior improvements in either metric. As there were also no differences between tSCS parameters, we selected 30Hz stimulation to tolerance for longitudinal training, for clinical feasibility reasons.

B. *Longitudinal Training Effects*

All four participants exhibited longer step lengths relative to baseline at various timepoints. SS09 and SS20 increased both sides' step length. SS10 and SS21 increased non-paretic step length – which are shorter than paretic steps – consistent with split-belt treadmill training interventions [4].

Step length symmetry (0-200 scale) showed minimal significant changes – only SS10 significantly improved step length symmetry. Of the two people who completed the three-month follow-up, one maintained symmetry improvements, while one reduced symmetry vs. baseline.

An open question in tSCS-augmented treadmill training paradigms is the number of sessions required to elicit improvement. Here, step length increases in all participants by the midpoint assessment, after 12 training sessions, informing future tSCS step length gait therapies. Step length symmetry shows no such clear cutoff.

C. *Relation Between Single-Day and Longitudinal Effects*

Recall that the four participants in the longitudinal training also participated in the single-day training. We hypothesized that the chronic stroke participants who were more responsive to single-day tSCS-augmented gait training (lower Avg. Stim vs. SHAM, Fig. 1A) may be more neuroplastic and therefore respond more readily to longitudinal tSCS. However, we have thus far not observed this relationship. The most consistent observation is that the longitudinal change from baseline is much larger than observed in a single session, as expected with a larger training volume.

D. *Limitations and Future Directions*

These longitudinal findings are necessarily preliminary, given the small sample size and current lack of a control group. We are actively collecting additional longitudinal data, including a control group. Future work will also explore the effects of tSCS on gait kinematics and muscle activity.

V. Conclusion

Transcutaneous spinal cord stimulation-augmented gait training is a promising neuromodulation technique to rehabilitate post-stroke gait. We observe minimal step length or step length symmetry differences after a single bout of tSCS-augmented treadmill training. By contrast, longitudinal training shows increased non-paretic or bilateral step length after 12 sessions, but step length symmetry improvement in one person only. We plan to expand this study to include a SHAM control group and larger sample size. Pending additional data, these preliminary findings suggest that tSCS-augmented treadmill training may be effective an effective tool in rehabilitating step length in people with chronic stroke.

References

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